YUHAN CHEN

Ph.D. Student | Computer Science and Engineering | University of Michigan

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RESEARCH INTERESTS

AI agents for scientific discovery; programmable laboratories; declarative experiment orchestration; intelligent workflows and resource scheduling; reliable research software systems.

EDUCATION

University of Michigan | Ann Arbor, MI — Aug. 2026 – Present

Ph.D. in Computer Science and Engineering

University of California, Berkeley | Berkeley, CA — Aug. 2020 – May 2024

B.A. in Computer Science and Data Science; Distinction in General Scholarship; GPA: 3.92/4.00; Dean's List (Spring 2022)

PUBLICATIONS

- Zhenning Yang, **Yuhan Chen**, Patrick Tser Jern Kon, Tongyuan Miao, Hongyi Lin, Venkat Viswanathan, Danai Koutra, and Ang Chen. "Experiment-as-Code Labs: A Declarative Stack for AI-Driven Scientific Discovery." *arXiv preprint arXiv:2605.04375 (2026)*. [arXiv]
- **Yuhan Chen**, Hongyi Lin, Tianyi Zhang, Adarsh Dave, Jared Mitchell, Jay Whitacre, and Venkatasubramanian Viswanathan. "ElyteOS: Software for Automated Electrolyte Preparation, Measurement, and Data Management." *SoftwareX 34, 102565 (2026)*. [DOI]
- Tianyi Zhang, Hongyi Lin, **Yuhan Chen**, and Venkatasubramanian Viswanathan. "SALSA: A Low-Cost Self-Driving Lab Modular Add-On for Salt Solubility Assessment for Battery Electrolytes." *Digital Discovery 5, 1881–1887 (2026)*. [DOI]

RESEARCH EXPERIENCE

Research Assistant | University of Michigan Jun. 2024 – Aug. 2026

Supervisor: Prof. Venkatasubramanian Viswanathan | AI-driven laboratory systems

- Developed ElyteOS, a Python-based automation and data-management platform that integrates heterogeneous instruments, experiment queues, persistent storage, and interactive monitoring and visualization.
- Designed optimization-based experiment preparation and resource allocation to improve throughput and reduce resource consumption; reported a 16× speedup and approximately 30% lower resource use in the evaluated workflow.
- Built LLM-driven multi-agent workflows for literature retrieval, experiment generation and validation, execution monitoring, and iterative planning based on experimental feedback.
- Contributed to Experiment-as-Code infrastructure for declarative procedure specification, resource-aware orchestration, and execution of instrument workflows using directed acyclic graphs.
- Developed a computer-vision pipeline for automated physical-state monitoring, combining YOLO-based region detection with image classification and temporal consistency checks.

Research Affiliate | Lawrence Berkeley National Laboratory / UC Berkeley Sep. 2022 – Dec. 2023

Supervisor: Prof. Gerbrand Ceder | Language-model methods for scientific information processing

- Developed a hybrid NLP pipeline combining language models and rule-based parsing to extract and classify structured attributes from more than 20,000 scientific terms.
- Evaluated few-shot prompting strategies, improving classification performance by 30–40% on selected categories; explored connecting literature-derived knowledge to automated research workflows.

TECHNICAL EXPERTISE

Programming: Python, C++, Java, SQL, JavaScript. **AI & learning:** LLM agents, retrieval-augmented generation, PyTorch, YOLO, reinforcement learning, Bayesian optimization. **Systems & optimization:** Git, SQLite, Pandas, NumPy, Plotly Dash, concurrent workflows, instrument communication (serial/HID), mixed-integer programming.